

Project RFM: Client Segmentation and Product Recommendation

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Summary

INTRODUCTION

1. EXPLORATORY DATA ANALYSIS

2. PRE-PROCESSING OF DATA AND RFM

3. ARTIFICIAL INTELLIGENCE MODEL

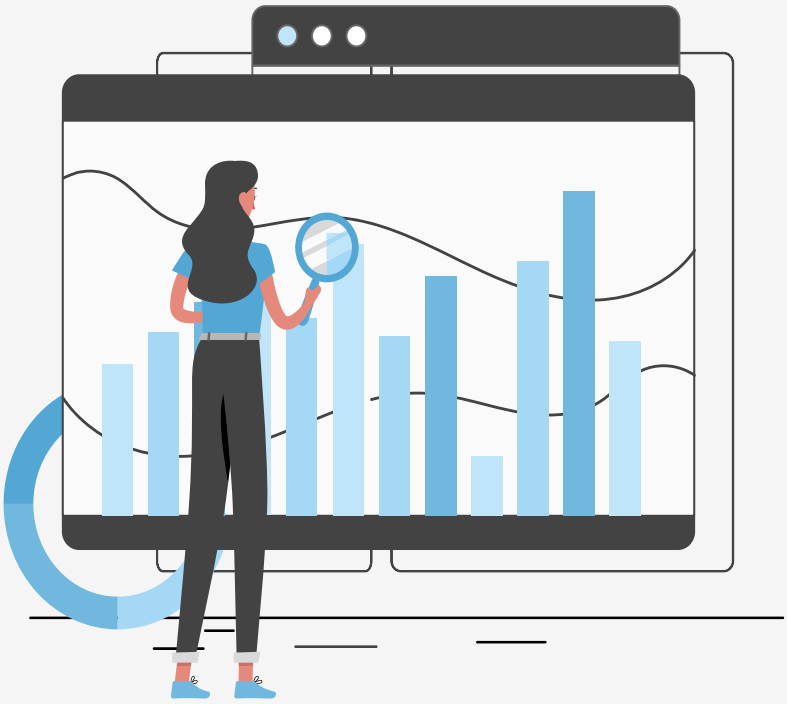
4. APPLICATION DEVELOPMENT

CONCLUSION

RESSOURCES

INTRODUCTION

Context



In retail, understanding customer behavior is essential for creating personalized experiences and optimizing marketing efforts. Using customer data, companies can identify buying habits, predict future purchases, and recommend products that match their preferences. Using machine learning and data analytics, companies can not only understand what customers are doing now, but also predict what they will do next, helping to improve customer satisfaction and loyalty.

As part of this analysis, we worked with a sales dataset to study customer behavior. By performing tasks such as customer segmentation, clustering, and analyzing key metrics (such as recency, frequency, and monetary value), We sought to gain insights that can be used to make data-driven business decisions. The goal is to provide practical advice for companies to improve their customer engagement strategies.

In this report, we will describe the steps we took to explore the dataset, apply clustering techniques, and implement a machine learning application that predicts and recommends products based on customer profiles. We will also discuss the technologies used in the development and implementation of the app, including how we integrated explainable components to help users better understand their personalized recommendations.

INTRODUCTION

Project

This project aims to use data analytics and machine learning techniques to understand customer behavior in a retail context. By analyzing transaction data, we seek to better understand customer purchasing habits, identify key metrics such as duration, frequency, and monetary value (RFM), and use clustering techniques to segment customers based on their purchasing habits. The goal is to develop a machine learning application that can predict customer behavior, recommend products, and provide a personalized shopping experience.



Plan of Action:

This project focuses on analyzing customer behavior in retail using machine learning and data analytics. The process will take place in several key stages:

- **Exploratory Data Analysis and Data Cleaning:** First, we will explore the dataset and prepare it for further use based on our needs and observations.
- **Customer Metric Calculation:** We will calculate important metrics such as recency, frequency, and monetary value (RFM) to understand customer behavior and identify high-value customers.
- **Clustering and Segmentation:** Using clustering algorithms (such as k-means), we will group customers based on their RFM scores to explore and segment customer profiles.
- **Explainable AI:** To improve transparency, we will integrate explainable AI capabilities, helping users understand the logic behind recommendations.

Predictive Application Implementation:

We will developed an application that predicts customer behavior and recommends products based on their RFM profile. The application will be made available online for easy access.

1. EXPLORATORY DATA ANALYSIS

Exploratory data analysis (EDA) is the first step in understanding data and preparing it for further analysis. As part of this project, we will perform EDA on the dataset to uncover insights and clean it for further use. Below are the main steps of the EDA process:

Loading and inspecting the data:

The first task is to load the dataset and inspect its structure. We will check the raw data to understand the data format and identify obvious problems, such as missing values or incorrect data types.

The dataset :

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850.0	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850.0	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850.0	United Kingdom
5	536365	22752	SET 7 BABUSHKA NESTING BOXES	2	2010-12-01 08:26:00	7.65	17850.0	United Kingdom

The columns are described as follows :

InvoiceNo	**Categorical**	**a 6-digit integral number uniquely assigned to each transaction. If this code starts with letter 'c', it indicates a cancellation**
StockCode	Categorical	a 5-digit integral number uniquely assigned to each distinct product
Description	Categorical	product name
Quantity	Integer	the quantities of each product (item) per transaction
InvoiceDate	Date	the day and time when each transaction was generated
UnitPrice	Continuous	product price per unit
CustomerID	Categorical	a 5-digit integral number uniquely assigned to each customer
Country	Categorical	the name of the country where each customer resides

Dataset informations :

```
df.shape
```

```
(541909, 8)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 541909 entries, 0 to 541908
Data columns (total 8 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   InvoiceNo    541909 non-null  object
 1   StockCode    541909 non-null  object
 2   Description  540455 non-null  object
 3   Quantity     541909 non-null  int64
 4   InvoiceDate  541909 non-null  datetime64[ns]
 5   UnitPrice    541909 non-null  float64
 6   CustomerID   406829 non-null  float64
 7   Country      541909 non-null  object
dtypes: datetime64[ns](1), float64(2), int64(1), object(4)
memory usage: 33.1+ MB
```

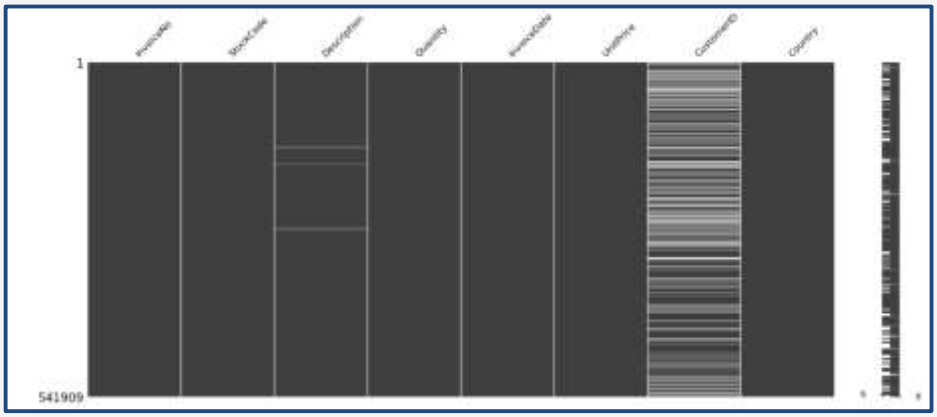
1. EXPLORATORY DATA ANALYSIS

Missing values :

The **missingno** matrix reveals a substantial number of missing values in our dataset. Notably, **24.93% of Customer ID entries are absent,** which requires careful consideration before taking any action. Simply dropping this column could lead to a significant loss of data, potentially skewing our analysis and resulting in biased outcomes.

After thorough investigation, **we concluded that imputing values for the missing Customer IDs was not the optimal solution**. The systematic nature of the missing data suggests a potential issue with data collection or management that should be addressed to improve future data quality. For the current analysis, **to avoid introducing further bias**, we have decided to proceed with deleting the rows with missing Customer IDs, despite the considerable data loss this entails.

```
Missing values in each column:
-----------------------------------
              Total  Percent
CustomerID   135080    24.93
Description    1454     0.27
InvoiceNo         0     0.00
StockCode         0     0.00
Quantity          0     0.00
InvoiceDate       0     0.00
UnitPrice         0     0.00
Country           0     0.00
```



After using .dropna() on the CustomerId column we can see the following stats :

```
# Let's have a look
retail.describe().T
```

	count	mean	std	min	25%	50%	75%	max
Quantity	406829.0	12.061303	248.693370	-80995.0	2.00	5.00	12.00	80995.0
UnitPrice	406829.0	3.460471	69.315162	0.0	1.25	1.95	3.75	38970.0
CustomerID	406829.0	15287.690570	1713.600303	12346.0	13953.00	15152.00	16791.00	18287.0

Then checking how many customers they actually have :

```
# The unique customers of the retailer

df.CustomerID.unique().shape
```

```
(4373,)
```

1. EXPLORATORY DATA ANALYSIS

EDA with Data Visualisation

We will start by analyzing each variable individually. We will analyze categorical variables by checking their distribution and unique values. And we will analyze continuous variables, by checking their summary statistics, distributions, and identifying outliers.

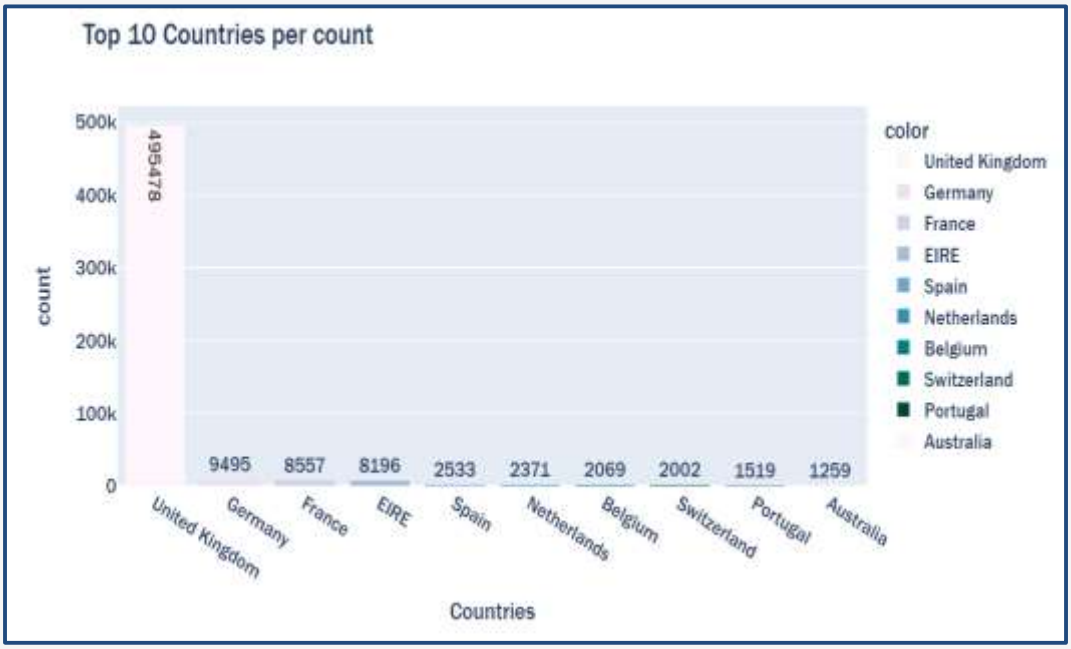
```
# Separate columns into numerical and categorical
numeric_features = retail.select_dtypes(exclude=['object']).columns
categorical_features = retail.select_dtypes(include=['object']).columns

# Print the results
print(f"We have {len(numeric_features)} numerical features: {list(numeric_features)}")
print(f"\nWe have {len(categorical_features)} categorical features: {list(categorical_features)}")
```

```
We have 4 numerical features: ['Quantity', 'InvoiceDate', 'UnitPrice', 'CustomerID']

We have 4 categorical features: ['InvoiceNo', 'StockCode', 'Description', 'Country']
```

Number of countries by number of customers count and Data Viz of the top 10 :




```
df['Country'].value_counts()
```

United Kingdom	495478
Germany	9495
France	8557
EIRE	8196
Spain	2533
Netherlands	2371
Belgium	2069
Switzerland	2002
Portugal	1519
Australia	1259
Norway	1086
Italy	803
Channel Islands	758
Finland	695
Cyprus	622
Sweden	462
Unspecified	446
Austria	401
Denmark	389
Japan	358
Poland	341
Israel	297
USA	291
Hong Kong	288
Singapore	229
Iceland	182
Canada	151
Greece	146
Malta	127
United Arab Emirates	68
European Community	61
RSA	58
Lebanon	45
Lithuania	35
Brazil	32
Czech Republic	30
Bahrain	19
Saudi Arabia	10

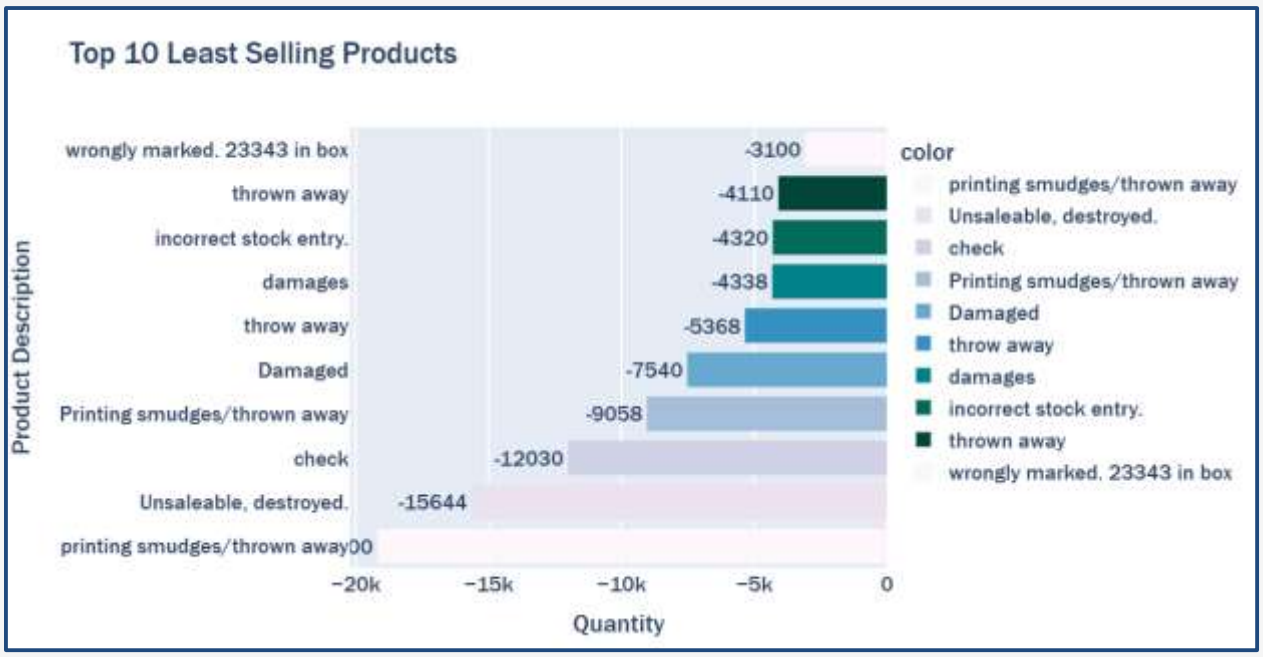
Name: Country, dtype: int64

Data Viz of the top 10 products by quantity :



1. EXPLORATORY DATA ANALYSIS

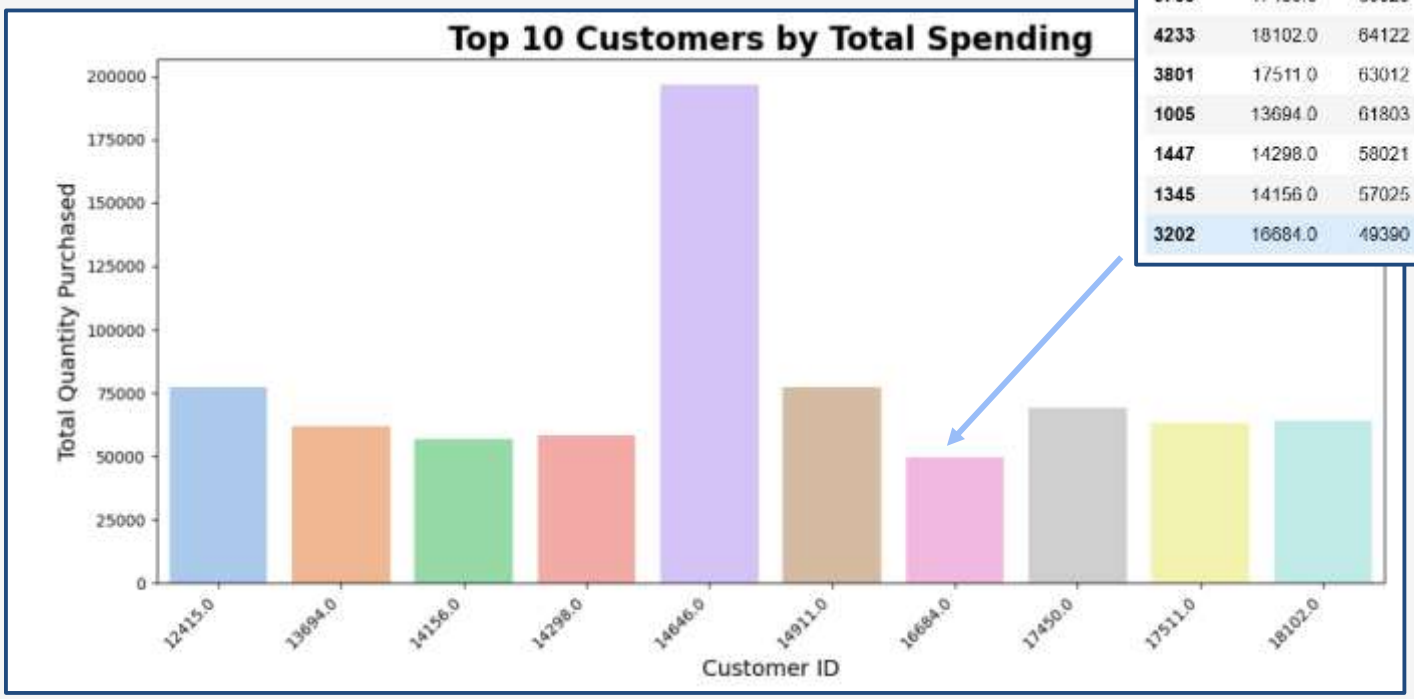
Data Viz of the top 10 least selling products by quantity :



There seem to be some damaged and not up-to-selling standard products in there. which means there may have been some return transactions in this data.

Data Viz of the top 10 Customers by Total Spending :

	CustomerID	Quantity
1703	14646.0	196719
55	12415.0	77242
1895	14911.0	77180
3758	17450.0	69029
4233	18102.0	64122
3801	17511.0	63012
1005	13694.0	61803
1447	14298.0	58021
1345	14156.0	57025
3202	16684.0	49390



Organizational methodology and work environment

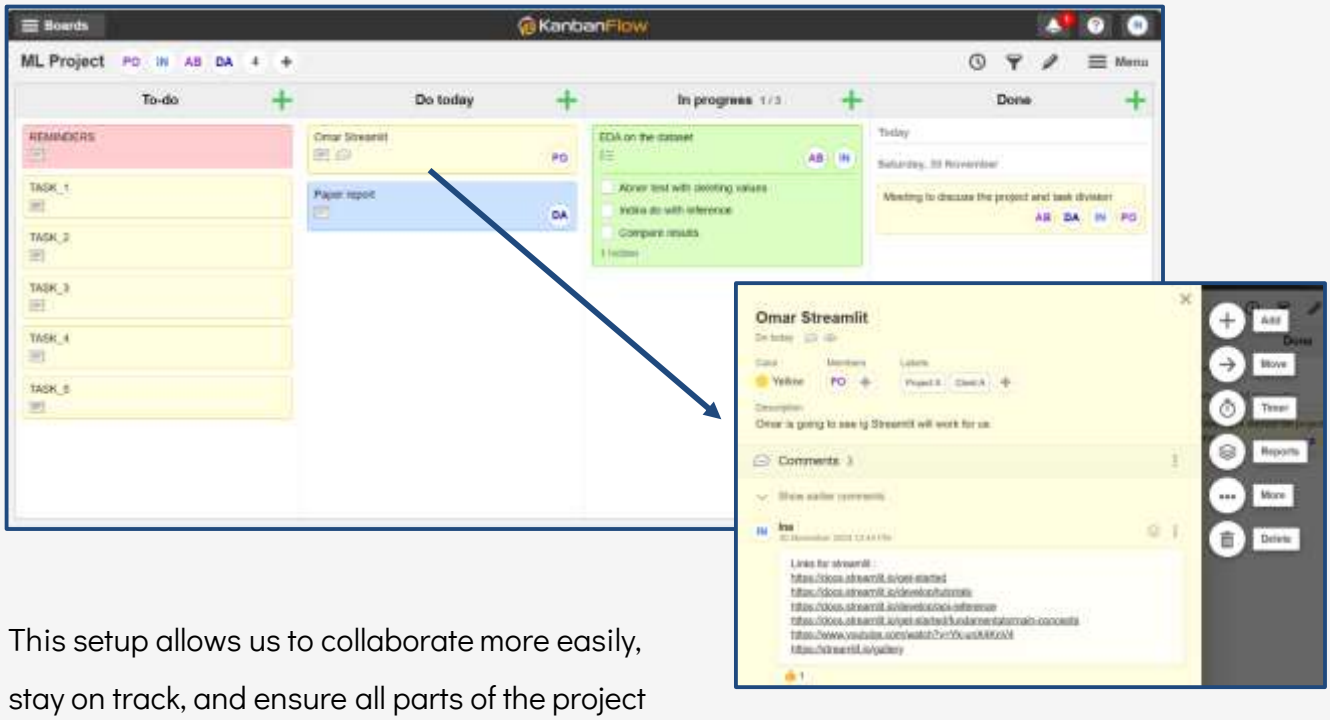
For this project, we established a structured and efficient work environment with clearly defined roles and responsibilities. **Omar serves as the Project Manager**, overseeing the final product (the application) and providing instructions to the team. **Abner and Indira, as Data Scientists,** focus on Exploratory Data Analysis (EDA) and model development. **Danish plays a versatile role, initially working on the report and later assisting Omar with the application development**.

We chose **Flask** as our web framework due to its reliability and scalability for application development. For analysis and model development, we utilize **Google Colab and Jupyter Notebook running on Windows Subsystem for Linux (WSL).** These tools provide the necessary computational resources for handling intensive tasks efficiently.

This combination of tools and technologies ensures smooth execution and testing throughout the project lifecycle, enabling us to meet the IT requirements and optimize the development process.

To stay on track and manage our time effectively, we adopted the **Agile** method using a tool named **KanbanFlow**. This method helps us break down tasks into focused chunks, which helps us stay on task longer and avoid distractions. By tracking the time spent on each task, we can identify areas that are taking longer than expected, which helps us adjust our approach and improve efficiency as we move forward with the project.

The management method used was as follows: **TO-DO, DO TODAY, IN PROGRESS and DONE**. We tag members that are in charge of the task and let them handle it their way. We shared links, code and insights by tagging the person on the task and putting it inside.



This setup allows us to collaborate more easily, stay on track, and ensure all parts of the project progress smoothly.

2. PRE-PROCESSING OF DATA and RFM

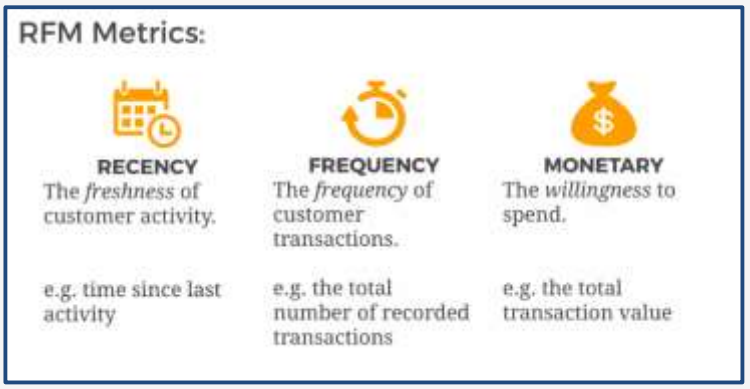
What Is RFM Analysis?

RFM stands for Recency, Frequency, and Monetary value. It is a method to analyze customer behavior and segment them into groups for better business decisions.

- Recency (R): How recently a customer made a purchase.
- Frequency (F): How often a customer makes purchases.
- Monetary Value (M): How much money a customer spends.

Why RFM Analysis?

1. Identify loyal customers.
2. Spot high-value customers.
3. Target lapsed customers with campaigns.



Data Preparation :

The first step is to organize customer transaction data into a table where each row represents a customer, and the columns are:

- Customer ID
- Recency: Days since the last purchase.
- Frequency: Total number of transactions.
- Monetary Value: Total amount spent

```
# Merge all RFM attributes (Monetary, Frequency, Recency) on CustomerID
rfm = pd.merge(rfm, rfm_r, on='CustomerID')

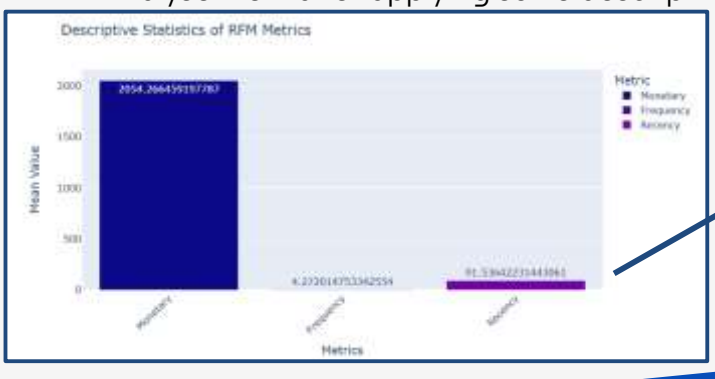
# Ensure column order and naming
rfm = rfm[['CustomerID', 'Monetary', 'Frequency', 'Recency']]

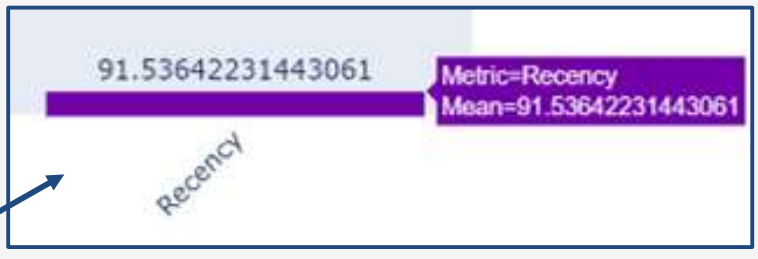
# Display the first few rows of the final RFM dataframe
print(rfm.head())
```

```
   CustomerID  Monetary  Frequency  Recency
0     12346.0  77183.60          1      325
1     12347.0   4310.00          7        1
2     12348.0   1797.24          4       74
3     12349.0   1757.55          1       18
4     12350.0    334.40          1      309
```

Since we don't have those values let's create them!

1. Import our transaction data.
2. Calculate:
   a. **Recency**: Find the difference between today's date and the last purchase date.
   b. **Frequency**: Count the number of transactions per customer.
   c. **Monetary Value**: Sum up the transaction amounts for each customer.
3. Merge all the RFM attributes into one dataset
4. Analyse them after applying some descriptive statistics.





2. PRE-PROCESSING OF DATA and RFM

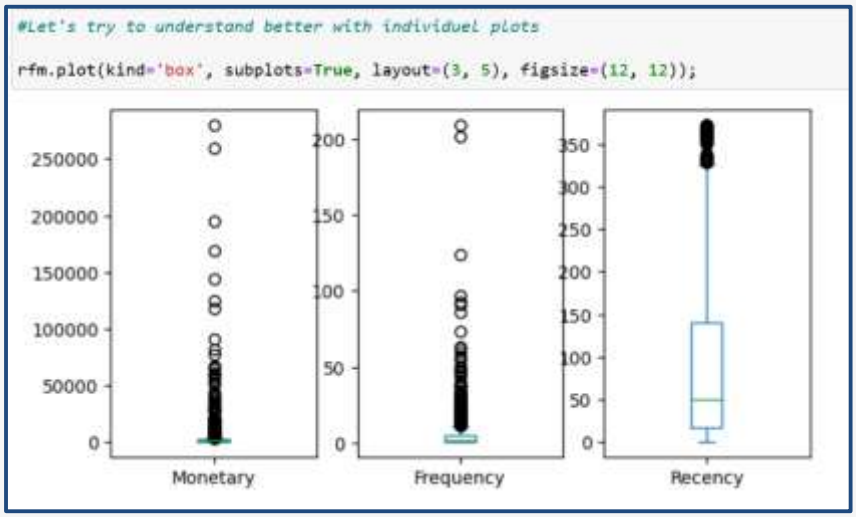
Data visualization of RFM

As a team, we believe that data visualization plays a key role in our data analysis process. It helps us to explain and understand the data before moving on to more complex tasks like training machine learning models. By visualizing the data, we can quickly spot anomalies or outliers, see how variables are distributed, and identify correlations between them.

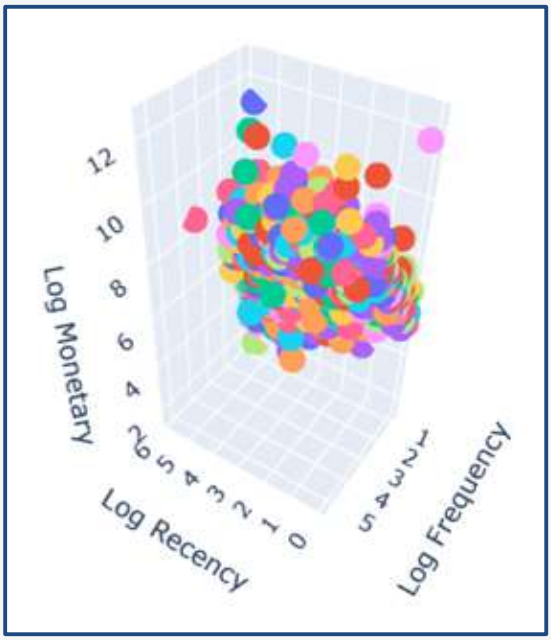
This step is crucial for choosing the right preprocessing techniques and evaluating their effectiveness. In addition, data visualization helps us present our findings to non-technical stakeholders, allowing them to better understand the results and make informed decisions. In short, visualization improves the overall quality and understanding of data, leading to more efficient and accurate models.

We first tried to understand those better with boxplots as seen below and with distplots :

```
#Let's try to understand better with individual plots
rfm.plot(kind='box', subplots=True, layout=(3, 5), figsize=(12, 12));
```



Then we decided to plot the customers RFM in a 3D plot where each axis is the log value of the previous metrics (after doing a log-transform on those metrics).



3D Plot of Customers by Log-Transformed RFM Metrics :

Final insights :

These insights are interesting, but they are presented individually. To gain a more comprehensive understanding, we need to combine them or develop a scoring system that allows us to visualize the data more effectively.

To effectively combine insights, we can create a scoring system that integrates both qualitative and quantitative data, allowing us to visualize and understand the information as a whole.

2. PRE-PROCESSING OF DATA and RFM

Scoring the RFM metrics:

We can calculate RFM (Recency, Frequency, and Monetary) scores in different ways, such as using percentiles, the 80/20 rule, or creating custom scores based on business needs. In this case, we will use percentiles for each RFM dimension.

To calculate percentiles, we sort customers based on a specific metric, divide them into equal groups, and assign labels to each group. For RFM, the labels will represent customer behavior:

- **Recency: from "Very Recent" to "Very Infrequent."**
- **Frequency: from "Very Frequent" to "Very Infrequent."**
- **Monetary: from "High Spending" to "Low Spending."**

After assigning labels for each RFM dimension, we combine them to create an overall RFM label for each customer. For example, if a customer has a Recency score of 4, Frequency of 5, and Monetary of 3, their combined label could be "Recent-Very Frequent-Average Spending."

This method gives priority to customers who are more recent, since recent engagement is usually more valuable to businesses. Higher scores for Frequency and Monetary values are also important, as companies want to encourage more visits and higher spending.

We divide each RFM dimension into five groups, each containing 20% of the customers. This approach is based on the distribution of customer data, shown through boxplots, and helps highlight differences in customer behavior. This method allows us to make better decisions by clearly showing how customers compare in terms of **recency, frequency, and spending.**

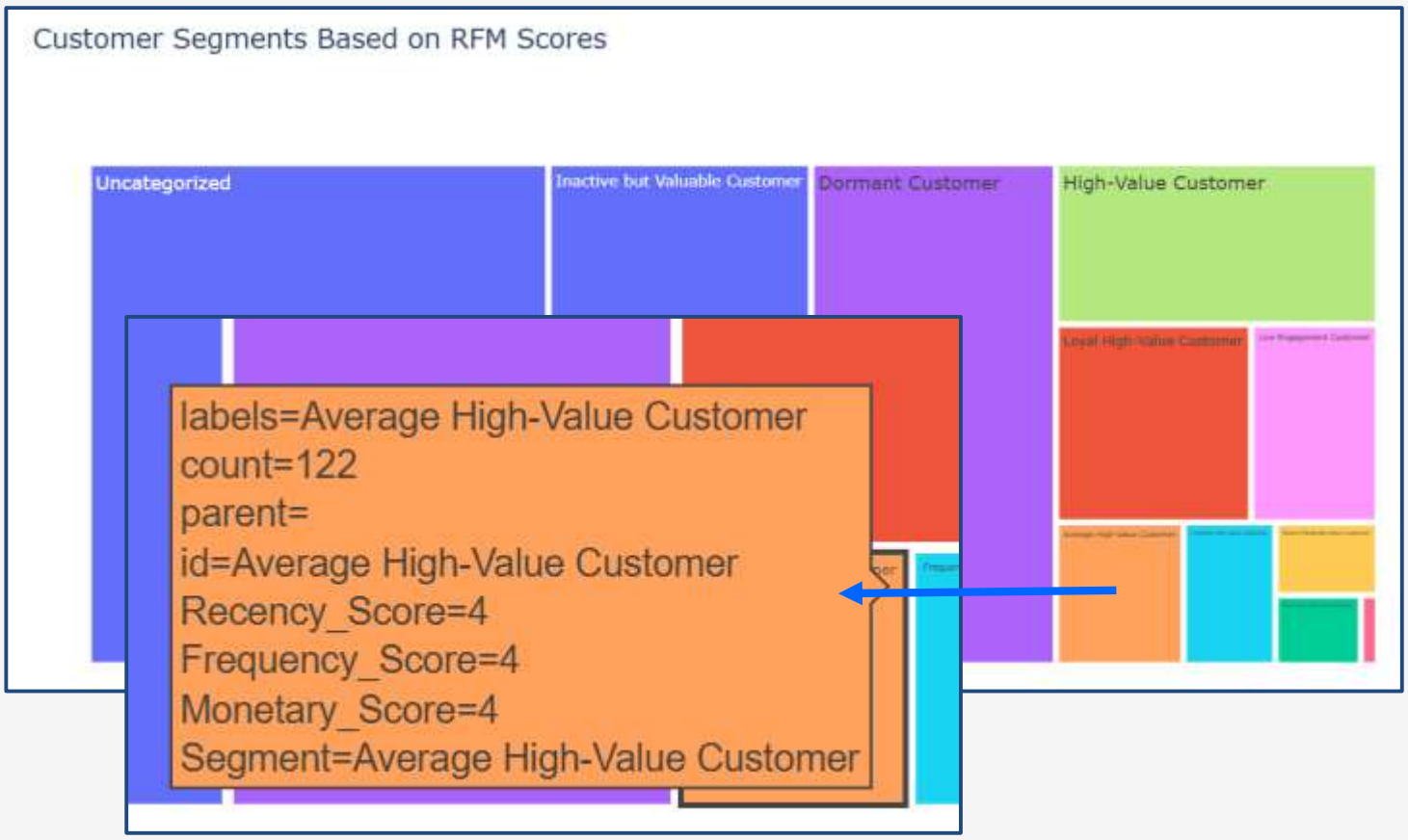
Final RFM scores dataset :

```
rfm_scores.head()
```

	CustomerID	Monetary	Frequency	Recency	Recency_Score	Frequency_Score	Monetary_Score
0	12346.0	77183.60	1	325	1	1	5
1	12347.0	4310.00	7	1	5	5	5
2	12348.0	1797.24	4	74	2	4	4
3	12349.0	1757.55	1	18	4	1	4
4	12350.0	334.40	1	309	1	1	2

2. PRE-PROCESSING OF DATA and RFM

Data Viz of Customer Segments based on RFM scores :



Final observation :

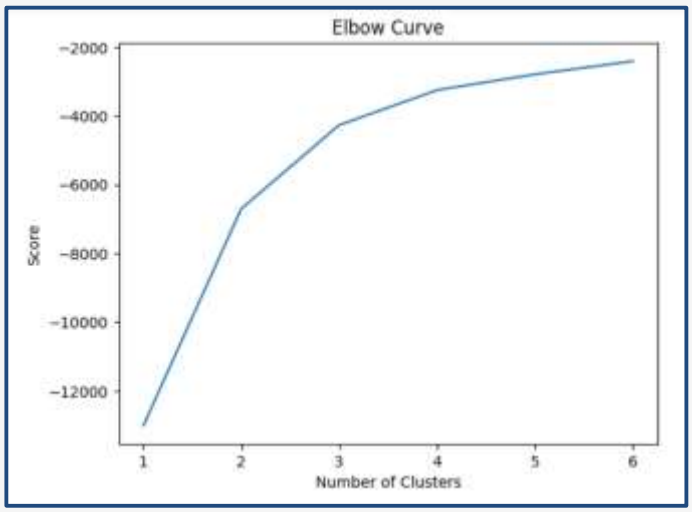
The current RFM segmentation method oversimplifies customer behavior by relying on arbitrary thresholds for recency, frequency, and monetary values. **This can lead to misrepresentations, such as categorizing a one-time high spender as a high-value customer.** Furthermore, this static approach fails to adapt to changes in customer behavior or future trends.

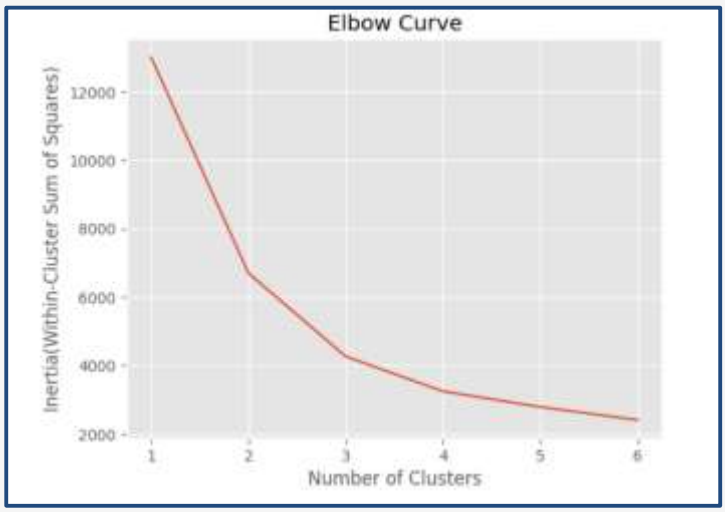
To gain a more accurate understanding of customer value and improve decision-making, we should **consider using K-means clustering**. This method allows for more granular segmentation and can dynamically adjust to customer behaviors, providing deeper insights into our customer base.

3. ARTIFICIAL INTELLIGENCE MODEL

Clustering with K-Means

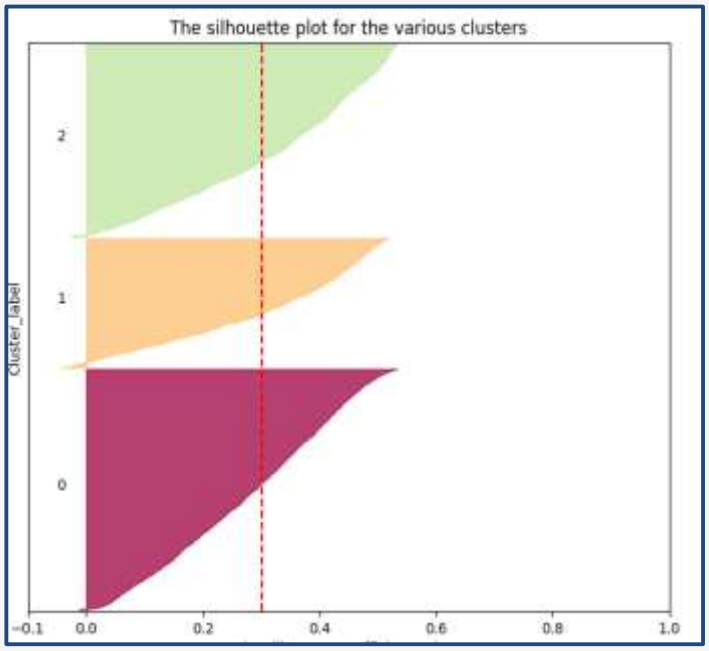
We decided to explore different approaches to chose our number of clusters like the Elbow curve

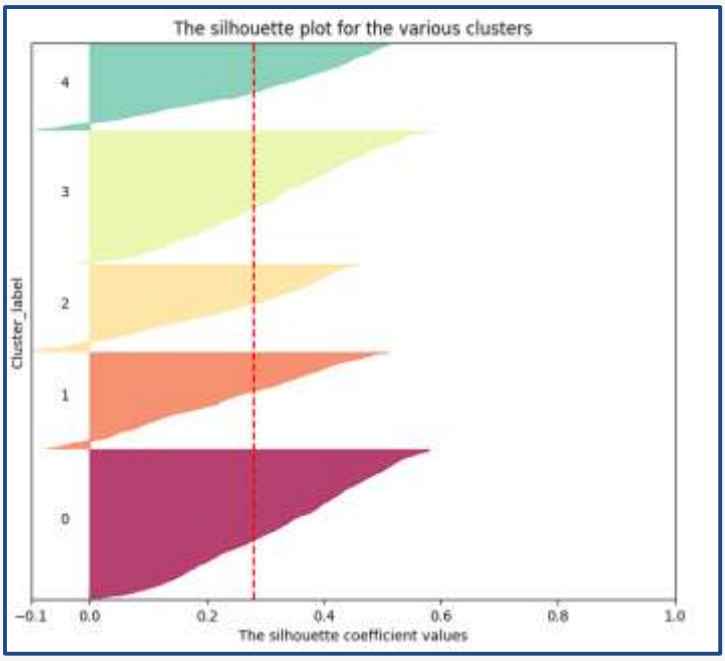




and the Silhouette.

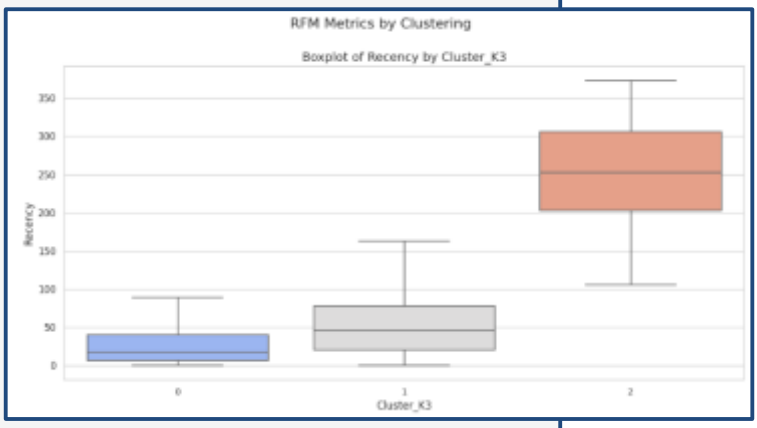
The best number of clusters seemed to be K=3 or K=5 so we decided to explore those two extensively.

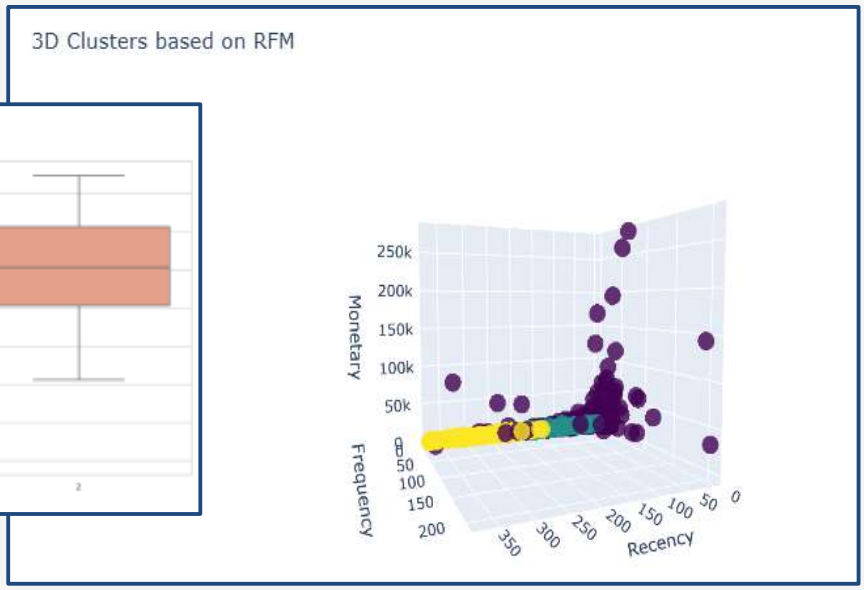




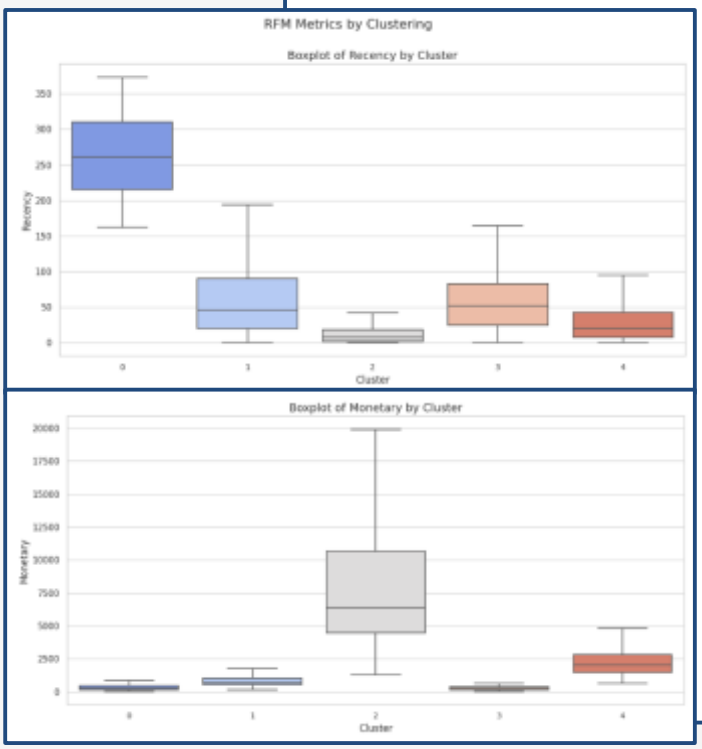
3. ARTIFICIAL INTELLIGENCE MODEL

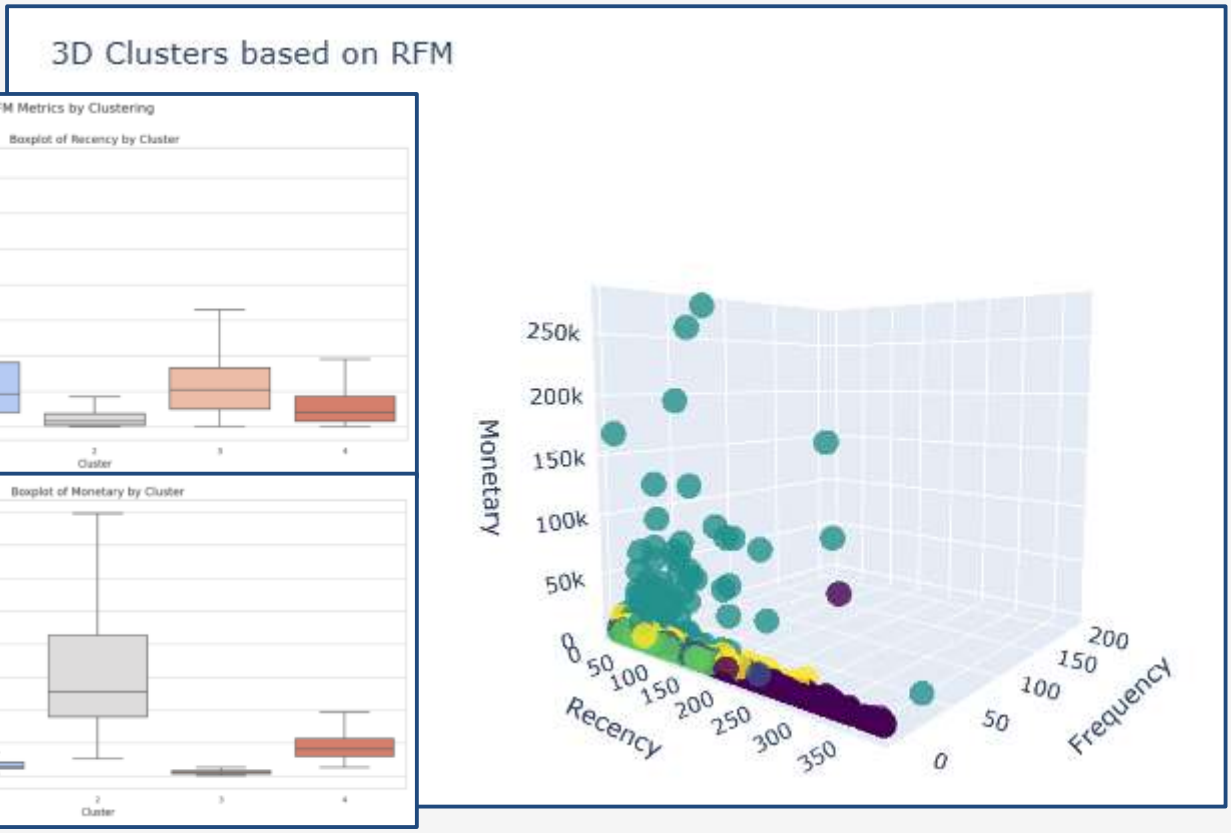
RFM metrics applied on the clusters of the K-Means K=3 :





RFM metrics applied on the clusters of the K-Means K=5 :





The **Model with 5 clusters seems to provide a more balanced segmentation that captures the diversity of customer behaviors**, from the most loyal, high-value customers to the least engaged ones. This strategy will enable us to personalize marketing and target recommendations tailored to each cluster's needs. By classifying customers into these distinct groups, we can create specific engagement strategies to increase customer retention, loyalty, and overall revenue.

3. ARTIFICIAL INTELLIGENCE MODEL

As part of the recommendation system, we used the 'all-MiniLM-L6-v2' model, which is a transformer-based model designed to map sentences and paragraphs into a 384-dimensional dense vector space and can be used for tasks such as clustering or semantic search.



We leveraged the cluster value and the products associated with the customers in each cluster. **We generated an embedding for each product description, with the aim of obtaining a representation of each cluster.**
This way, we could identify products that were close to their centroid. For each cluster centroid, we calculated the **cosine similarity with the embeddings of each product to find the top 10 most similar products.**
However, the result were not as expected, as **the top 3 recommended products were quite similar** across different clusters.

	StockCode	Cluster	Description	Cleaned_Description
0	23166	0	MEDIUM CERAMIC TOP STORAGE JAR	medium ceramic top storage jar
1	85116	4	BLACK CANDELABRA T-LIGHT HOLDER	black candelabra tlight holder
2	22375	4	AIRLINE BAG VINTAGE JET SET BROWN	airline bag vintage jet set brown
3	71477	4	COLOUR GLASS. STAR T-LIGHT HOLDER	colour glass star tlight holder
4	22492	4	MINI PAINT SET VINTAGE	mini paint set vintage

This could be due to the fact that **products from different clusters may have high similarity with the centroids of other clusters.**
This phenomenon could also occur because, although the products have different descriptions in terms of exact words, the **SentenceTransformers** model can generate representations with similar semantic contexts.

Example of top 5 when the customer is in the cluster 0

	StockCode	Description
79080	35095B	RED VICTORIAN FABRIC OVAL BOX
231287	35095A	BLUE VICTORIAN FABRIC OVAL BOX
91882	21195	PINK HONEYCOMB PAPER BALL
117126	90125A	PINK BERTIE GLASS BEAD BAG CHARM
117124	90125D	PURPLE BERTIE GLASS BEAD BAG CHARM

An improvement could have been to try another sentence-transformers model more specific to the product domain or to use a weighting approach when calculating similarities.

4. APPLICATION DEVELOPMENT

For our final application we decided to use the 5 clusters described in ourK-Means model which is based on customer behavior, that was segmented according to the **Recency**, **Frequency**, and **Monetary** (RFM) scores. Each cluster represents a different category of customer engagement and value. Here's a breakdown of the **5 clusters** we've created:

1. High-Value Customers (Cluster 0)
Description: These are the customers who are the **most valuable** to the business. They have high **Recency**, **Frequency**, and **Monetary** scores.

- **Profile**:
  - These customers have made recent purchases.
  - They buy frequently and spend a significant amount.

They are the **loyal** and **profitable** customers who generate the most revenue.

2. Moderate-Value Customers (Cluster 1)
Description: These customers are valuable, but not as consistently engaged as **high-value** customers. They might have a **moderate** recency score or a moderate level of spending.

- **Profile**:
  - Customers who have bought more than once but aren't buying as frequently.
  - They may spend a good amount when they do make a purchase, but they aren't as active as high-value customers.

They could be targeted with **special offers** to increase engagement.

3. Dormant or Low-Engagement Customers (Cluster 2)
Description: These are customers who have been **inactive** or have very low engagement with the business.

- **Profile**:
  - These customers have **low Recency**, meaning they haven't bought in a while.
  - Their **Frequency** and **Monetary** scores are low, indicating that they don't engage much or don't spend much when they do purchase.

These customers might need re-engagement strategies, such as **retargeting** ads or **personalized offers** to win them back.

4. Inactive but Valuable Customers (Cluster 3)
Description: These customers have **low recency** but have previously shown **high frequency** or **high monetary** value. They are not currently engaged, but they have the potential to be valuable again.

- **Profile**:
  - These customers have purchased frequently or spent large amounts, but they haven't bought recently.

They could be considered **high potential targets for reactivation**. They may need a specific outreach strategy, like **re-engagement campaigns** offering discounts or new products

5. Churned Moderate-Value Customers (Cluster 4)
Description: These customers were once moderately valuable but have **stopped buying** or **show low engagement**. They have a moderate monetary or frequency score but a **low recency** score.

- **Profile**:
  - These customers were more engaged in the past but have recently gone inactive.
  - They might still be valuable if re-engaged correctly, but they are at risk of **churn**.

They might need to be targeted with **retargeting ads** or **loyalty rewards** to bring them back.

4. APPLICATION DEVELOPMENT

Summary of the 5 Clusters:

1. **High-Value Customers**: Recent, frequent, and high spenders — the most profitable group.
2. **Moderate-Value Customers**: Good spenders but not as frequent or recent. They represent potential growth.
3. **Dormant or Low-Engagement Customers**: Low interaction and purchase frequency. These customers are inactive and need re-engagement.
4. **Inactive but Valuable Customers**: Low recent activity but have historically shown value. They can be reactivated.
5. **Churned Moderate-Value Customers**: Moderate previous value but now dormant. They represent a risk of churn.

Why 5 Clusters?

These 5 clusters provide a **balanced** segmentation that captures the diversity of customer behaviors, from the most loyal, high-value customers to the least engaged ones. The strategy enables **personalized marketing** and **targeted recommendations** tailored to each cluster's needs. By classifying customers into these distinct groups, you can create specific **engagement strategies** to increase customer retention, loyalty, and overall revenue.

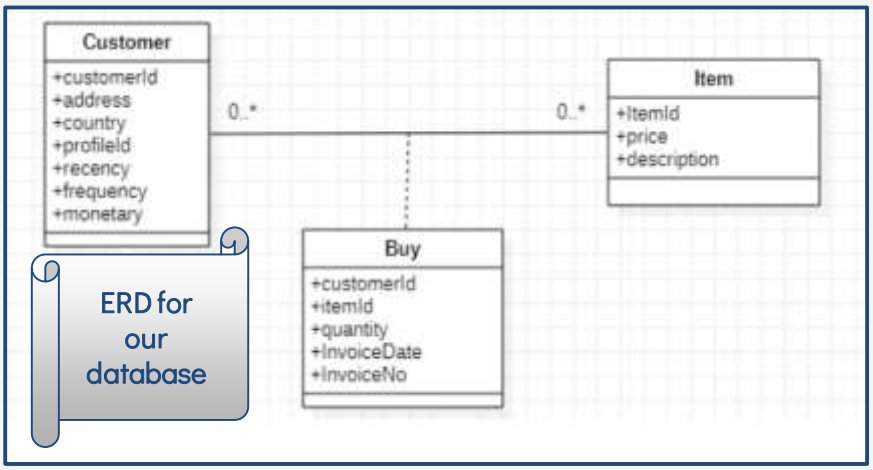
Our App and How the Recommendation System Works

The goal of the product recommendation application is to suggest products to customers based on their purchasing behavior, classified into **Recency**, **Frequency**, and **Monetary (RFM)** categories. This is done by grouping customers into clusters and offering recommendations for products that similar customers have purchased.

Process Flow:

1. **Customer Input**: A customer connects on our application system and then he chose a list of **products** they are interested in.
2. **Backend Processing**:
   - Customer Profile Lookup: The system uses the Customer ID to retrieve the RFM score of the customer and their cluster (a group of customers with similar purchasing behaviors).
   - Recommendation Generation: Based on the cluster the customer belongs to, the system retrieves the most popular products bought by other customers in the same group. It ensures that products the customer has already expressed interest in are excluded from the recommendations.





4. APPLICATION DEVELOPMENT

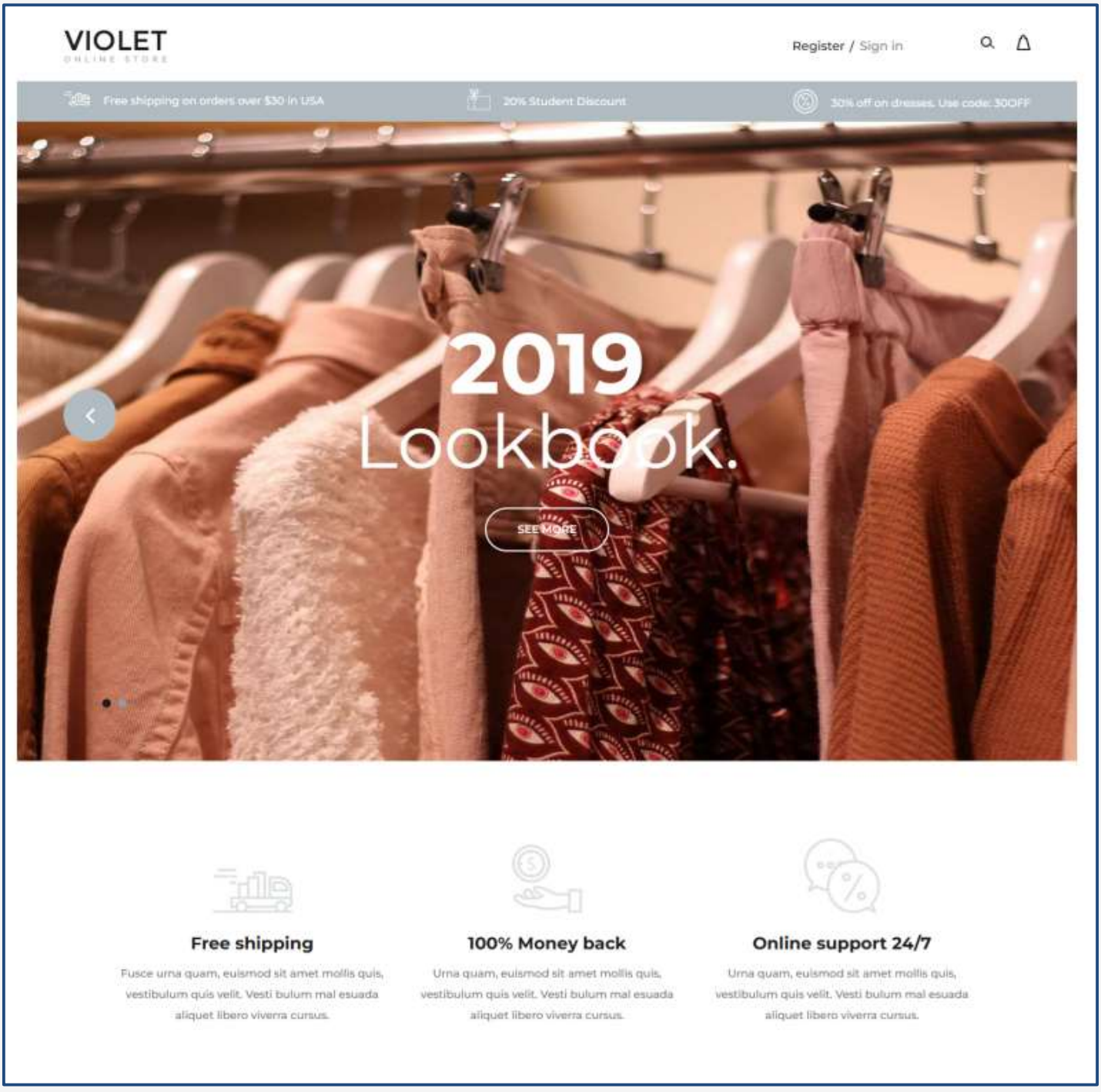
3. Recommendations Display: The backend returns two main outputs:

- The Customer's RFM Values (Recency, Frequency, and Monetary scores), which provide insight into the customer's purchasing behavior.
- A list of Recommended Products, based on the behaviors of similar customers.

4. Frontend Display: The customer sees:

- Their RFM profile (a description of their purchasing behavior).
- A list of products they may be interested in.

The welcome page of the application :



4. APPLICATION DEVELOPMENT



Pros and Cons of the System

Pros:

- Personalized recommendations based on customer behavior
- Data-driven insights using RFM analysis
- Scalable approach for larger datasets
- Simple implementation without complex machine learning models

Cons:

- Highly dependent on data quality and completeness
- Cold start problem for new customers
- Lack of real-time preference adaptation based on new products

Explanation Alignment with the Code

The code implements RFM-based clustering and product recommendation by:

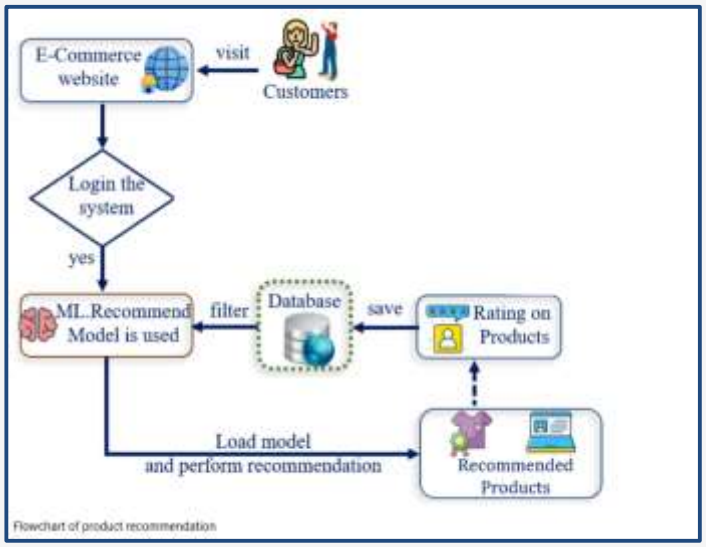
1. Handling RFM scores and transaction data
2. Clustering customers based on RFM scores
3. Recommending products based on cluster-specific frequent purchases

Code Efficiency

Areas for improvement:

- Optimize cluster-based recommendation process
- Enhance non-buyer identification efficiency
- Consider distributed computation for better scalability





Function Design

The generate_cluster_recommendations function:

- Groups similar customers
- Handles non-buyers
- Excludes products of existing interest
- Provides relevant, personalized recommendations

Web Application Integration

- Backend: Flask API processes customer data and generates recommendations
- Frontend: HTML form for customer input
- Pickle: Used for model persistence

The RFM-based system effectively personalizes product recommendations. While there's room for optimization, it provides a simple, scalable solution for tailoring suggestions based on customer behavior.

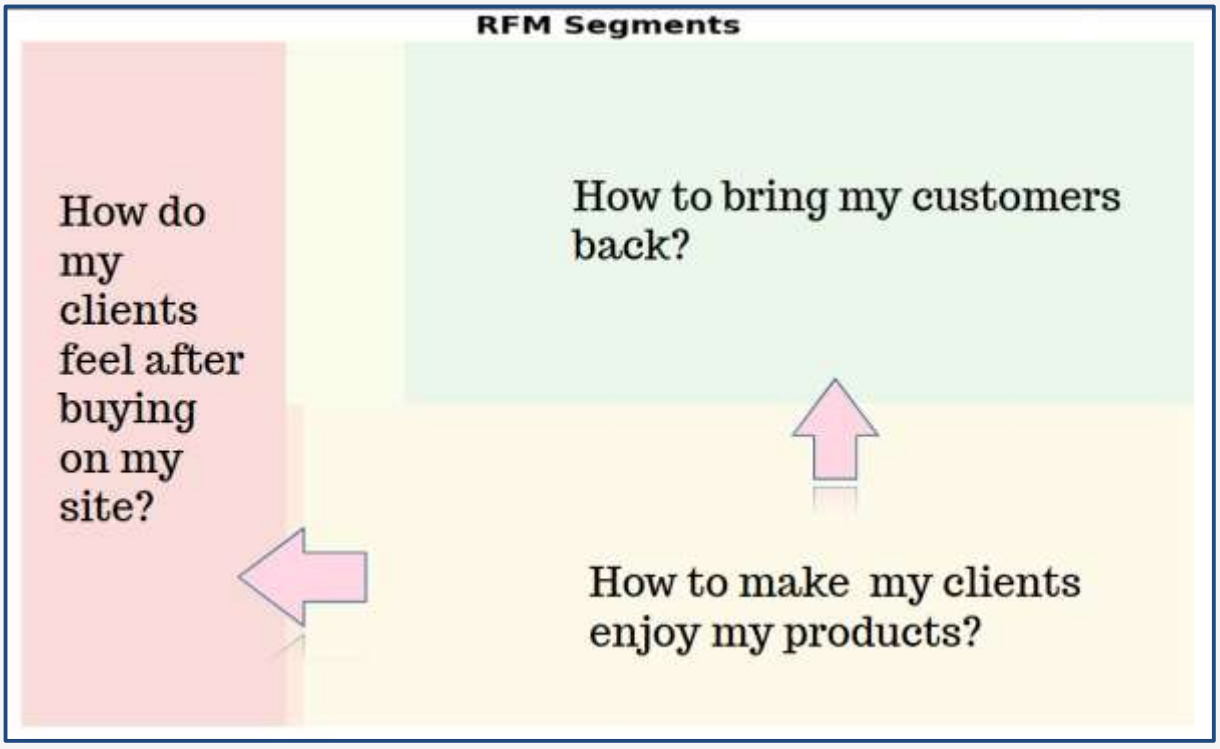
CONCLUSION

In conclusion, our RFM-based product recommendation system represents a powerful and innovative approach to personalized marketing. By leveraging customer purchasing behavior and advanced clustering techniques, we've created a solution that not only enhances the shopping experience but also drives business growth.

Key achievements of our system include:

1. Precise customer segmentation into 5 distinct clusters, enabling targeted marketing strategies
2. Personalized product recommendations that significantly increase the likelihood of purchase
3. A scalable and efficient design that can handle large datasets and integrate seamlessly with existing web platforms
4. Clear, actionable insights for marketing teams to tailor their efforts and re-engage customers effectively

While there's room for future optimizations, particularly in non-buyer identification and large-scale performance, the current system already provides a robust foundation for data-driven decision-making and customer-centric marketing.



By implementing this RFM-based approach, businesses can expect to see improved customer satisfaction, increased sales, and more efficient marketing spend. This project demonstrates the power of data analytics in transforming customer relationships and driving tangible business outcomes in the competitive e-commerce landscape.

Ressources

RFM Segmentation | PPT

https://towardsdatascience.com/recency-frequency-monetary-model-with-python-and-how-sephora-uses-it-to-optimize-their-google-d6a0707c5f17

https://guillaume-martin.github.io/rfm-segmentation-with-python.html

https://medium.com/@ilaydakursun/customer-segmentation-with-rfm-analysis-learn-more-about-your-customers-f12b348acbd

https://towardsdatascience.com/find-your-best-customers-with-customer-segmentation-in-python-61d602f9eee6

https://towardsdatascience.com/a-gentle-introduction-on-market-basket-analysis-association-rules-fa4b986a40ce

Flowchart of product recommendation | Download Scientific Diagram

https://www.slideshare.net/slideshow/rfm-segmentation/46901188